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Windows PowerShell
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W10D2_LangGraph_State_Machines> notepad langgraph_state_machine.py
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W10D2_LangGraph_State_Machines> python langgraph_state_machine.py

=====
TEST 1: TECHNICAL INPUT
=====
[CLASSIFY] Input: How do I reset my password?
[CLASSIFY] Classification: technical
[ROUTE] Selected route: technical_response
[RESPONSE] This is a technical request. Please verify the technical details and follow the appropriate system or application instructions.

Final State:
{'user_input': 'How do I reset my password?', 'classification': 'technical', 'route': 'technical_response', 'response': 'This is a technical request. Please verify the technical details and follow the appropriate system or application instructions.'}

=====
TEST 2: GENERAL INPUT
=====
[CLASSIFY] Input: What is artificial intelligence?
[CLASSIFY] Classification: general
[ROUTE] Selected route: general_response
[RESPONSE] This is a general information request. The system can provide a general explanation based on the user's question.

Final State:
{'user_input': 'What is artificial intelligence?', 'classification': 'general', 'route': 'general_response', 'response': 'This is a general information request. The system can provide a general explanation based on the user's question.'}

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TEST 3: HUMAN-IN-THE-LOOP
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[CLASSIFY] Input: Should I share my medical information online?
[CLASSIFY] Classification: sensitive
[ROUTE] Selected route: human_review
[HUMAN REVIEW] Graph execution paused.
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Windows PowerShell

Graph paused for human review.
Interrupt detected:
[Interrupt(value={'message': 'Human review required for this sensitive request.', 'input': 'Should I share my medical information online?', 'question': 'Approve this request? Enter yes/no or provide guidance.', id='78d364a156693dc6d0d278260edf01d1'})]

[HUMAN REVIEW] Graph execution paused.
[HUMAN REVIEW] Received: yes
[RESPONSE] Human review approved the request. The system can proceed while following appropriate privacy and safety guidelines.

Graph resumed successfully.

Final State:
{'user_input': 'Should I share my medical information online?', 'classification': 'sensitive', 'route': 'human_review', 'response': 'Human review approved the request. The system can proceed while following appropriate privacy and safety guidelines.', 'human_feedback': 'yes'}

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TEST 4: TECHNICAL INPUT
=====
[CLASSIFY] Input: How can I connect Python to a database?
[CLASSIFY] Classification: technical
[ROUTE] Selected route: technical_response
[RESPONSE] This is a technical request. Please verify the technical details and follow the appropriate system or application instructions.

Final State:
{'user_input': 'How can I connect Python to a database?', 'classification': 'technical', 'route': 'technical_response', 'response': 'This is a technical request. Please verify the technical details and follow the appropriate system or application instructions.'}

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TEST 5: GENERAL INPUT
=====
[CLASSIFY] Input: What is machine learning?
[CLASSIFY] Classification: general
[ROUTE] Selected route: general_response
[RESPONSE] This is a general information request. The system can provide a general explanation based on the user's question.
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Windows PowerShell
{'user_input': 'Should I share my medical information online?', 'classification': 'sensitive', 'route': 'human_review', 'response': 'Human review approved the request. The system can proceed while following appropriate privacy and safety guidelines.', 'human_feedback': 'yes'}

=====
TEST 4: TECHNICAL INPUT
=====
[CLASSIFY] Input: How can I connect Python to a database?
[CLASSIFY] Classification: technical
[ROUTE] Selected route: technical_response
[RESPONSE] This is a technical request. Please verify the technical details and follow the appropriate system or application instructions.

Final State:
{'user_input': 'How can I connect Python to a database?', 'classification': 'technical', 'route': 'technical_response', 'response': 'This is a technical request. Please verify the technical details and follow the appropriate system or application instructions.'}

=====
TEST 5: GENERAL INPUT
=====
[CLASSIFY] Input: What is machine learning?
[CLASSIFY] Classification: general
[ROUTE] Selected route: general_response
[RESPONSE] This is a general information request. The system can provide a general explanation based on the user's question.

Final State:
{'user_input': 'What is machine learning?', 'classification': 'general', 'route': 'general_response', 'response': "This is a general information request. The system can provide a general explanation based on the user's question."}

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W10D2 TESTING COMPLETED
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All 5 inputs were tested successfully.
Conditional routing was verified.
Human-in-the-loop pause and resume was verified.
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